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CODE:CSA0910-JAVA

1. Write a program to store 15 integer values in an ArrayList and remove all duplicate elements without using another collection

RemoveDuplicates.java	Output
<pre>1 import java.util.ArrayList; 2 import java.util.Scanner; 3 4 public class RemoveDuplicates { 5 public static void main(String[] args) { 6 Scanner sc = new Scanner(System.in); 7 ArrayList<Integer> list = new ArrayList<>(); 8 9 System.out.println("Enter 15 integers:"); 10 11 for (int i = 0; i < 15; i++) { 12 list.add(sc.nextInt()); 13 } 14 } 15 }</pre>	<pre>Enter 15 integers: 1122 11221 112211 112211 112211 112211 112211 112211 212 22222</pre>

2. Store employee IDs in a LinkedList and display them in both forward and reverse order.

EmployeeIDs.java	Output
<pre>1 import java.util.LinkedList; 2 import java.util.Scanner; 3 4 public class EmployeeIDs { 5 public static void main(String[] args) { 6 Scanner sc = new Scanner(System.in); 7 LinkedList<Integer> ids = new LinkedList<>(); 8 9 System.out.print("Enter number of employee IDs: "); 10 int n = sc.nextInt(); 11 12 System.out.println("Enter Employee IDs:"); 13 for (int i = 0; i < n; i++) { 14 ids.add(sc.nextInt()); 15 } 16 } 17 }</pre>	<pre>Enter number of employee IDs: 123 Enter Employee IDs: 6534</pre>

3. Store email IDs entered by the user in a HashSet. Display unique email IDs and count them.

```
UniqueEmails.java
7      HashSet<String> emails = new HashSet<>();
8
9      System.out.print("Enter number of email IDs: ");
10     int n = sc.nextInt();
11     sc.nextLine();
12
13     System.out.println("Enter Email IDs:");
14     for (int i = 0; i < n; i++) {
15         emails.add(sc.nextLine());
16     }
17
18     System.out.println("Unique Email IDs:");
19     for (String email : emails) {
20         System.out.println(email);
    }
```

Output

Enter number of email IDs: 32541

4. Count the frequency of each word in a paragraph using a HashMap

```
WordFrequency.java
14
15     for (String word : words) {
16         word = word.replaceAll("[^a-zA-Z]", "");
17
18         if (!word.isEmpty()) {
19             if (map.containsKey(word)) {
20                 map.put(word, map.get(word) + 1);
21             } else {
22                 map.put(word, 1);
23             }
24         }
25     }
26
27     System.out.println("Word Frequencies:");
28     for (String key : map.keySet()) {
29         System.out.println(key + " : " + map.get(key));
    }
```

Output

Enter a paragraph:
gdhJFYGdkbhfsjYwTFDIUMBfHJwHSfDKWYSTdGKHJDBCN
Word Frequencies:
gdhjfygdkbhfsjywtfdiumbfhjwhsfdkwystdgkhjdbcn : 1

=== Code Execution Successful ===